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Unsteady Aerodynamics of Rotorcraft in Ground Effect *

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This paper brings together experimental and computational work done in the study of low speed ground effect of rotorcraft. The paper recaps previous work done experimentally to study the unsteadiness in the rotorcraft flowfield using flow visualization and hot wire measurements. These experimental results are compared with computational results obtained using a classical vortex method coupled with an image system to simulate the ground. The comparisons are adequate; however, the computations use a lifting-line approach leading to some discrepancies in time traces of velocities in the wake. The paper also presents experimental results obtained in measuring fuselage loads in ground effect.

I. Introduction

The flowfield around a rotorcraft flying close to the ground presents a fluid dynamics problem where large fluctuations are observed under nominally quasi-steady flight conditions, separated by seemingly-random intervals. In ground effect (IGE) conditions, the wake of a helicopter rotor interacts with the ground which causes significant perturbation to the flow near the rotor blades, as well as the rest of the craft. Interactions between the main rotor wake and the ground have been associated with the formation and passage of a ground vortex in transitional flight.

The basic question that arises from the reported flight test results, is whether

1. The unsteadiness arises from the craft interacting with different regions of an otherwise quasi-steady flowfield, due to changes in wind direction, ground clearance or aircraft speed;

2. Or whether long-period fluctuations are generated in an otherwise periodic flowfield under fixed flight conditions.

Detailed experimental studies of the rotor wake and ground vortex were performed with an isolated model rotor above a static ground plane at low advance ratio, and various ground heights. This study aims to quantify time scales of unsteadiness by bringing together results from pulsed laser sheet flow visualization, hotwire anemometry, and fuselage force measurement. In previous work with this experimental set-up, it was shown that the wake was steady enough in the absence of ground effect, to enable clear quantification of the unsteadiness caused by ground effect. This unsteadiness was quantified using laser sheet imaging of vortex dynamics. It was then shown that large transient velocity fluctuations occurred with long intervals, in the ground-vortex and in the rotor inflow regions. The present paper investigates a decision point in the investigation of unsteady ground effect. It is argued that transients could occur due to two basically different situations, or a combination of these situations

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a) Due to wind direction and advance ratio change, so that the fuselage/tail rotor interacts with different parts of the rotor wake. However, the rotor wake for a given flight condition is perfectly periodic. This situation can be modeled analytically through quasi-steady methods.

b) Even though the flight condition remains identical, interactions between vortices, the ground and the vehicle surfaces, generate transient spikes in the forces and moments, with large intervals possible between such occurrences. This situation is fundamentally unsteady, and requires long-period unsteady simulation methods with high accuracy.

In actual flight tests, these situations may be indistinguishable, and may both occur simultaneously.

This paper brings together computational results with experimental measurements to support this assertion. Experiments have captured both these situations- the unsteady long time fluctuations in the hotwire measurements and quasi-steady fuselage force variation due to wake structure and its interaction with the fuselage and tail rotor. The next requirement is to facilitate incorporation of these phenomena in handling qualities codes. To achieve this, reliable computational simulation of these effects in all observed operating conditions are necessary. The paper attempts to show that the long time scale unsteadiness in the flowfield seen in hotwire measurements can be modeled using long time vortex simulations of the rotor flowfield in ground effect. The paper also presents the quasi-steady results obtained through fuselage force measurements. In the computations, the presence of the ground is modeled using the method of images. A lifting-line theory is used to model each rotor blade and a time-stepping vortex method has been used to obtain the evolution of tip-vortex. A numerical scheme has been designed to compute up to a hundred rotor revolutions. The computational results agree quite well with the experiments qualitatively but there are quantitative differences.

II. Experimental Background

The interaction of a rotorcraft wake with the ground during low speed flight close to the ground involves two distinct flow regimes:[1]

1. Recirculation of the wake ahead of the rotor at very low advance ratios, causing additional inflow through the forward part of the rotor; the Recirculation Regime.

2. Formation of a horseshoe ground vortex under the rotor at higher advance ratios with its associated interactions; the Ground Vortex Regime.

Studies have found irregular changes in hub moment for a helicopter flying close to the ground. Curtiss *et al.*[1] noted that the flow fluctuates at a low frequency in the recirculation regime. The authors in Reference 2 found that in the recirculation flow regime, there is very little lateral flow and moderate level of unsteadiness in the flow field. The unsteady phenomena have been attributed to the periodic interference fluctuation between downwash and upwash in the recirculation regime.[3]

When rotors are operating near the ground at low advanced ratios, the forward part of rotor wake, after impinging on the ground plane, flows forward and then rolls up, forming a recirculation zone around the rotors. For a single rotor, the ground vortex center in the longitudinal symmetric plane is a little upstream of rotor leading edge.[4]

Above a critical advance ratio, which depends upon collective pitch and height-to-diameter ratio, a well-defined concentrated horseshoe vortex is formed under the rotor, and significant lateral flow is present. The flow field becomes quite steady. The vortex becomes smaller, and eventually vanishes as the advance ratio is increased. The ground vortex also alters the helicopter yaw control effectiveness in sideway and rearward flights near the ground. These regimes of flight are shown in Figure 1.

The ground vortex itself is quite unsteady, both in size and shape, and also in location.[5] The ground vortex has a large-scale low-frequency pulsating behavior, which is referred to as puffing of the ground vortex. The sequence of this puffing behavior is as follows First the ground vortex is very small, but growing. For slow forward speed or for hover in light winds, the ground vortex would stay upstream. As the wind velocity increases, the ground vortex moves closer to the rotor. Interaction between the old vortices and new vortices increases as forward airspeed increases. As

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the ground vortex continues to grow, it eventually becomes too large for the flow field to sustain. At this point the entire flow field breaks up violently, and the large ground vortex is swept downstream. Immediately a new small ground vortex begins to grow upstream, and the cyclic process repeats itself.[6]

Tip vortex geometry is changed close to the ground since the wake contracts and then expands after impinging on the ground, leading to change in radial trajectory of the tip vortex when compared to flight away from the ground.[7] The unsteadiness is also more in the radial direction.

For a helicopter going into forward flight, the required power increases in the 5-20kt speed regime. The change in inflow through the rotor not only affects the required power but the lateral trim as well. Under OGE conditions at low speed the down flow through the front part of the rotor is lower than that through the rear part. This phenomenon makes the rotor tend to flap with the retreating side up. The effect of the ground vortex is to increase the down flow through the front part of the rotor disc, making the flow more uniform and reducing the requirement for left stick. After the ground vortex is overrun, left stick is required suddenly.[8]

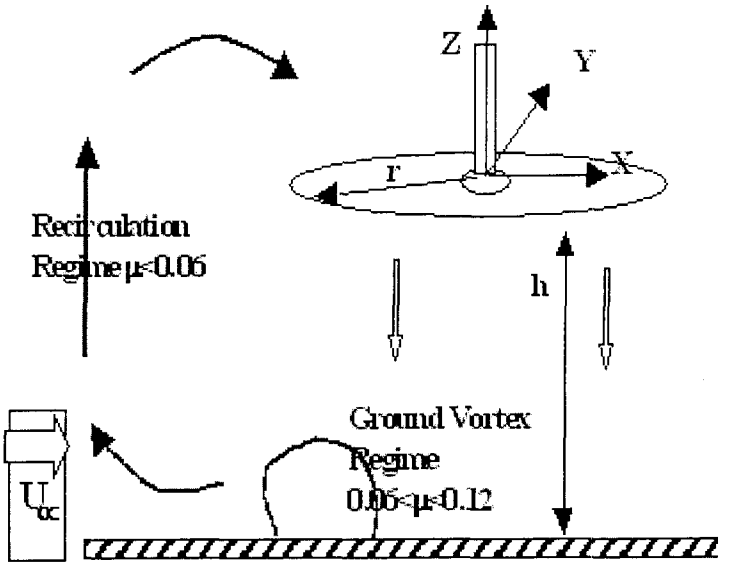


Figure 1. Rotor coordinate system and flight regimes.

When a helicopter is in rearward or sideward flight, the ground vortex may occur behind or on the side of the vehicle and may affect the tail rotor. In some helicopter development, the ground vortex causes the loss of tail rotor thrust.[9]

III. Experimental Description

The experiments were conducted in the John J. Harper Low Speed Wind Tunnel of Georgia Institute of Technology. This facility is a closed-circuit wind tunnel with a 7ft × 9ft test section. A DC motor is used to drive the rotor shaft at speeds between 0 and 2100 rpm. The specifications of the test rotor are shown in table 1.

Diameter(ft)	3
Chord length	3.37
Number of blades	2(Teetering)
Collective pitch angle(deg)	10(fixed)
Twist angle(deg)	0
Rotor shaft tilt angle(deg)	6
Solidity	0.1193

Table 1. Test Rotor Specifications

Previous experimental work qualitatively and quantitatively studied the unsteadiness of the flowfield around a

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rotor in ground effect. Flow visualization and hot-wire measurements were done with and without a fuselage below the rotor. Fluctuation frequencies were studied at various locations in the flowfield using hot-wire probes aligned with their sensors horizontal and perpendicular to the freestream.

In previous work with this set-up,[10] it was shown that the wake was steady enough in the absence of ground effect, to enable clear quantification of the unsteadiness caused by ground effect. This unsteadiness was quantified using laser sheet imaging of vortex dynamics. It was then shown that large transient velocity fluctuations occurred with long intervals, in the ground-vortex and in the rotor inflow regions. At $0.03 < \mu < 0.04$, the rotor tip vortices moved forward and upward, recirculating through the rotor disk. Seven or eight vortices were tracked in the loop. The time scale of recirculation was approximately 0.11secs (frequency 9Hz) in this case. Recirculation was not observed even at low advance ratios in the OGE case.

To understand the evolution of the ground vortex, flow visualization images with gradually accelerating freestream flow (from $\mu = 0.058$ to 0.067 in 45 secs) were captured. At $\mu = 0.056$, separation occurred but a ground vortex was not formed yet. A t $\mu = 0.058$, the separated flow rolled and formed a ground vortex. I n the range $0.060 < \mu < 0.067$, the ground vortex moves downstream as increased. It was observed that the separation point was moving back and forth intermittently at a fixed μ. As μ is increased, the mean location of the separation point moved downstream and the amplitude of the oscillatory movement became smaller.

Flowfield velocity fluctuation time scales were studied using hot-wire anemometry, with the wire oriented horizontally along the y-direction, perpendicular to the freestream and vertical components. Figure 2 shows that the fluctuation amplitude is greatly increased in ground effect over the OGE case, reaching 10% of freestream speed. The IGE case also shows large spikes separated by long time scales. Lines corresponding to 2 standard deviations are drawn on the figure to identify the large transient spikes. The frequency content of the velocity fluctuations is dominated by 2 per rev fluctuations (blade passage), but the IGE case also shows large low-frequency content.

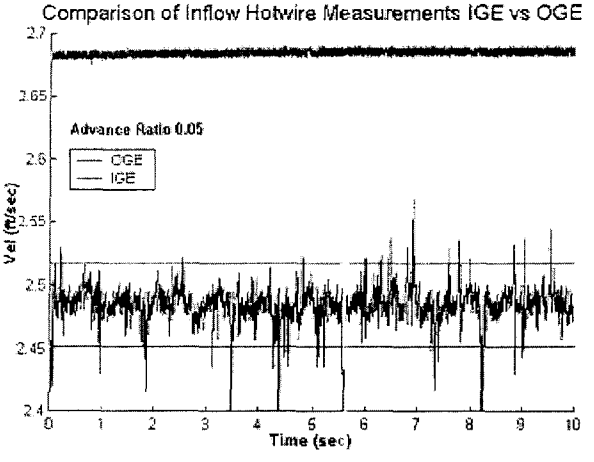


Figure 2. Hotwire Measurements Inflow IGE vs OGE.

The results of the hotwire measurements close to the ground vortex showed that ahead of the ground vortex a velocity fluctuation at low frequency (1 to 2 Hz) occurs and appears to be related to the movement of the flow separation point ahead of the nascent ground vortex.

The time history of velocity measurements IGE at an advance ratio of 0.05 is shown in Figure 3a. A high level of unsteadiness with a large time scale was seen for IGE case, while the flow field for OGE was seen to be much steadier. The nature of fluctuation can be seen in the figure, and it can be correlated with the flow structure seen in the flow visualization picture. Quantitatively, in the IGE case, we see blips occurring on the scale of a second ahead of the ground vortex, which seems to be related to the fluctuation of the separation point ahead of the ground vortex. The flow field was also plotted in the Fourier space (Figure 3b) for the same conditions and this helps in identifying the peak frequency of fluctuation in the flow field. It can be seen that there is low frequency domination in the flowfield ahead of the ground vortex. These low frequency effects are also seen in the hotwire measurements in the inflow plane suggesting that there is an effect of this fluctuation on thrust produced by the rotor.

The isolated-rotor experiments served to establish that

- we could indeed achieve clean periodic flow conditions in the absence of ground effect.
- that the rise in unsteadiness due to ground effect was clearly perceptible and quantifiable.
- that wake interaction with the ground does cause unsteadiness in the ground vortex.

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- that recirculation of vortices does cause fluctuations in the rotor inflow plane.
- long-period unsteadiness is indeed seen in the flowfield.

IV. IGE Wake Model and Numerical Scheme

The presence of a ground plane below a helicopter rotor introduces the requirement that the velocity component normal to the ground be zero. Because the fluid flow is assumed inviscid and incompressible, employing the Method of Images automatically satisfies this boundary condition. The model is depicted in Figure 4.

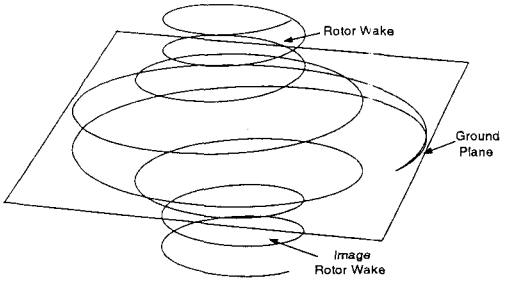


Figure 4. Model for In-ground Effect.

An orthogonal right-handed non-rotating system of coordinates is used here as shown in Figure 5. A single bladed rotor is being considered. The blade moves in the x-y plane and the z-axis points along the direction of downwash. The blade is assumed to be rigid and is represented by a lifting line. The azimuth ψ, (not shown in figure) is measured from the positive y-axis in the counter clockwise sense.

The rotor wake is assumed to consist of only the tip vortices and the influence of the inboard sheet and the root vortex are neglected. In the numerical scheme, the tip-vortex is composed of straight line vortex segments of equal strength and the initial condition employed is a smoothly contracting helical assembly of vortices. The following equations which describe an expanding helix set up the initial wake geometry for a rotor in ground effect

$$\begin{aligned} z^0_{vj,r} &= \bar{W}(j\Delta\psi) \\ x^0_{vj,r} &= f_e \times sin(\Theta - j\Delta\psi) \\ y^0_{vj,r} &= f_e \times sin(\Theta - j\Delta\psi) \end{aligned} \tag{1}$$

where the wake expansion factor, f_e is initially assumed to be equal to $1.0 + 2 \times j/N_p$ based on initial trial runs as suggested by Kini.[11] $\Delta\psi$ is the spatial step size. Initially for blades 1 and 2, $\Theta = 0$ and π respectively. The superscript '0' refers to the position at time t=0 and ($x^0_{vj,r}$, $y^0_{vj,r}$, $z^0_{vj,r}$) are made dimensionless by the rotor radius. $\bar{W}$ is the average downwash at the rotor disk non-dimensionalised by the rotor tip speed ΩR. Here, $j = 0, ..., N_p$ is the total number of advance points in the real wake. $x^0_{vj,r}$, $y^0_{vj,r}$, $z^0_{vj,r}$ are the coordinates of the vortex nodes in the real wake.

We assume an inviscid model in which the vorticity evolves as material elements of the fluid, convecting with the local fluid velocity in a Lagrangian sense. At any field point the velocity induced by the tip-vortex is calculated using the Biot-Savart law with a smoothing parameter given by,

$$\vec{U}_V(\vec{X},t) = -\frac{\Gamma}{4\pi}\int_C \frac{(\vec{X}-\vec{X}')\times d\vec{X}'}{(|\vec{X}-\vec{X}'|^2+\mu_c^2)^{3/2}} \tag{2}$$

where $\vec{X}$ = (x,y,z) is the field point and $\vec{X}'$ = (x',y',z') the position vector of a point on the vortex and C denotes the space curve occupied by the tip-vortices. All lengths in equation (2) are non-dimensionalised by the rotor radius R and the velocity components are non-dimensionalised by $\frac{\Omega R}{2\pi}$, where Ω is the angular velocity of the blade. The time is non-dimensionalised by $\frac{\Omega}{2\pi}$. Thus the dimensionless circulation is $\Gamma = 2\pi\Gamma^*/\Omega R^2$ where Γ^* is the dimensional

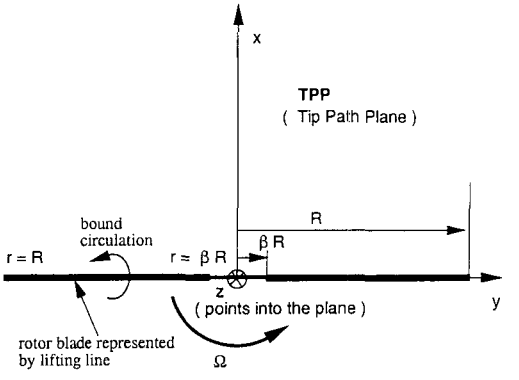


Figure 5. Co-ordinate system.

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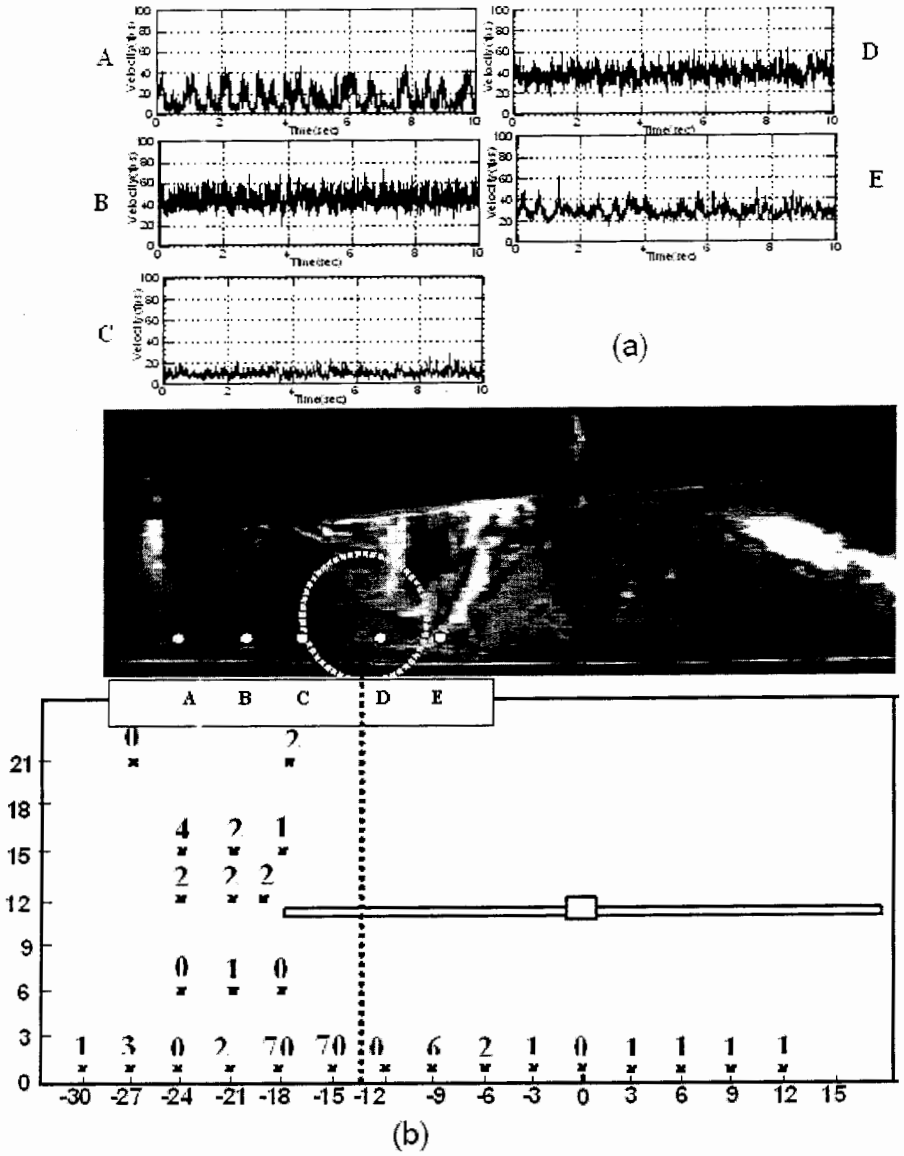


Figure 3. (a) Hotwire velocity trace at advance ratio 0.05 IGE, Points A, B, C, D, E marked in picture. (b) Peak Frequency of hotwire measurements at advance ratio 0.05 IGE. X and Y axes are in inches

circulation. The flow in the vortex is assumed to be a Rankine vortex with radius a_v. The smoothing parameter based on a Rankine core has been shown by Affes *et al.*[12] to be $\mu_c = a_v e^{-3/4}$.

The bound circulation of a rotor blade increases gradually from the root to near the tip where it reaches the maximum. It drops to zero at the tip over a very short distance.[13] The trailing vortices between the maximum bound circulation and the blade tip roll over to form the tip vortex. The other trailing vortices form the inboard sheet. Thus, to compute the tip vortex strength the following equation has been used.

$$\Gamma_{tip} = \int_0^{y_{max}} \frac{d\Gamma_b}{dy} dy = \Gamma_{b,max} \tag{3}$$

where y_{max} is the distance from the tip of the blade to the maximum bound circulation location. Thus the tip vortex circulation is equal to the maximum bound circulation.[14] The maximum bound circulation is obtained from the formula given in Stepniewski and Keys[15] given by

$$\Gamma_{b,max} = \frac{m}{2}\alpha(1 + \mu sin\psi_b - W) \tag{4}$$

where m is the lift curve slope, α the angle of attack, μ the advance ratio and W the downwash near the blade tip.

Each point on the tip-vortex is advanced in time by solving

$$\frac{d\vec{X}_v}{dt} = \vec{U} \tag{5}$$

where $\vec{X}_v$ is the position vector of a point on the vortex and $\vec{U}$ is the total velocity induced at that point. Thus $\vec{U}$ is the sum of the velocity induced by the helical tip-vortex, $\vec{U}_V$ (Equation (2)), the velocity induced by the bound circulation of each blade, $\vec{U}_{blade}$ and the non-dimensional forward flight speed along the positive $y - axis$, V_f. Thus,

$$\vec{U} = \vec{U}_V + \vec{U}_{blade} + \vec{U}_{V,i} + \vec{U}_{blade,i} + 2\pi V_f \tag{6}$$

where the subscript i stands for the image wake velocity contribution. Here the advance ratio is $\mu = V_f$.

While carrying out the computations, it is not necessary to calculate the velocities induced at the nodes in the image wake. The velocities induced by both the wakes are computed at all nodes in the real wake alone and these points are then stepped in time to their new positions in the next time step. The new image wake is then obtained as a mirror image of this new real wake.

The solution is stepped in time using an Adams-Moulton scheme with a Runge-Kutta starting formula. The integral on the right hand side of equation (5) is evaluated at advance points, while the integration of the first order system of ODE equation (5) is carried out using integration points. The integration points are located at the mid-point of each straight-line vortex segment.[11] After every time step, each of the advance points is moved to its new location. This method is observed to give better results than where the advance points and integration points coincide.[16] An initial wake geometry has to be specified for the time-marching scheme which is then allowed to evolve in time. The initial geometry of the wake is taken to be helical corresponding to a prescribed wake.

In this scheme, the vortex segment closest to the ground is removed after every time step and a segment is added near the blade. The contribution of this removed segment to the velocity of any point in the wake will be very small since its effect is almost cancelled by its image vortex segment below the ground. To model the removed points,since the tip vortices possess a nearly circular structure near the ground, a vortex ring is introduced close to the ground. The strength of the vortex ring is equal to the tip circulation at each time step. Each tip vortex is represented by 1600 nodes based on numerical accuracy issues. In forward flight the position of the ring is shifted slightly in the direction of the inflow. Thus a fixed number of points can be used and this saves a lot of computational time. Additional details can be found in Pulla and Conlisk.[17]

The accuracy of the numerical scheme was verified by varying the initial spatial step size and time step from $\Delta\psi = 8^o$ down to $\Delta\psi = 1^o$. We find that the spatial step $\Delta\psi = 2^o$and time step $\Delta t = 2^o$ insures accuracy up to three decimal places in the computed vortex trajectories.

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V. Comparison with Experiments

Results have been obtained for a two bladed rotor above the ground at a non-dimensional height 0.72. The velocities obtained from the computations have been compared with the results obtained from the experiments. Two points have been taken,one near the ground and the other in the middle of the wake as shown in Figure 6. From the experiments only the total velocity can be obtained. Individual components in the x, y and z directions cannot be calculated. Hence, for the sake of comparison with experiments, the net magnitude of the total velocity has been computed using the equation

$$v_{net} = \sqrt{u^2 + v^2 + w^2}$$

We have taken the velocity values obtained between 50 and 150 revolutions to compare with the experiments. Figure 6 shows the locations of the points P1 and P2 where the velocities have been computed. Figure 6 shows the comparison between the velocities obtained by the computations and those obtained from the experiments at point P1. The magnitude of the velocities obtained from the experiments and computations are comparable but there are differences in the behavior of these velocities. There is some similarity in the velocity profiles from the experiments and the computations between 2 and 3 seconds. The two peaks between 2 and 2.5 sec in Figure 6(a) correspond to the two peaks in the same time period in Figure 6(b). In the same way, the velocities between 2.5 and 3 seconds look quite similar. However it has to be noted that in both the cases the velocity is quite unsteady. The average value of velocity over 100 revolutions in the experiments is 6.029 ft/sec while that in the computations is 5.72 ft/sec.

Figure 7 shows the velocity comparisons for a point near the ground. The magnitude of the velocities is considerably higher in this case. The average over 100 revolutions in the experiments is 14.6 ft/sec while the corresponding value in the computations is 13.85 ft/sec. Even here the velocities are very unsteady but the corresponding peak values of velocities between 2 and 3 seconds seem to match pretty well. Hence, the computations are able to predict the velocities quantitatively to a reasonable accuracy.

VI. Fuselage Side Force Measurements

The next step in the experimental plan was to measure the fuselage forces. This was necessary to verify if the unsteadiness in the flow structure affects the forces on the fuselage. This experiment would also provide information about the quasi-steady forces on the fuselage.

A. Experimental Setup

The fuselage was mounted through a hole in the ground plane, and was set up to yaw about an axis that was perpendicular to the ground. A drawing of the experimental setup is shown in Fig.8. The fuselage is a fiberglass cylinder 34" long, with a hemispherical nose with a radius of 5.5".Positive side force is in the +y direction as shown in Figures 1 and 8. Positive yaw of the fuselage is clockwise when viewed from above as can be seen from Figure 8.

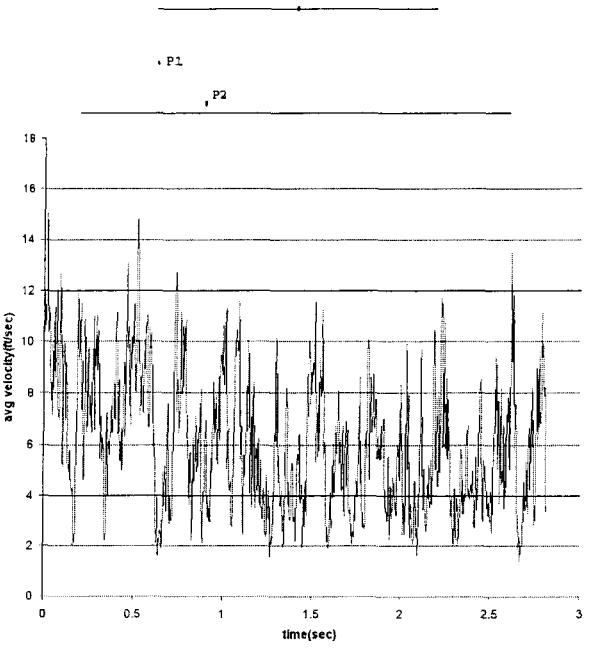
B. Summary of Experimental Results

Detailed fuselage force results were presented in an earlier paper.[18] The side force on the fuselage as a function of μ at zero yaw is shown in Figure 9. Fourier analysis of the flucuations showed 2-per-rev and 1-per-rev energy OGE, but in IGE, there was also a substantial increase of fluctuations at the system natural frequencies - which were determined using a step response test. The low frequency fluctuations seen in hotwire measurements are not discernible in the fuselage force measurements. This is to be expected, because the hotwire fluctuations seen in the inflow and ground vortex are localized fluctuations and their effects tend to be smoothed out when integrated over the entire fuselage.

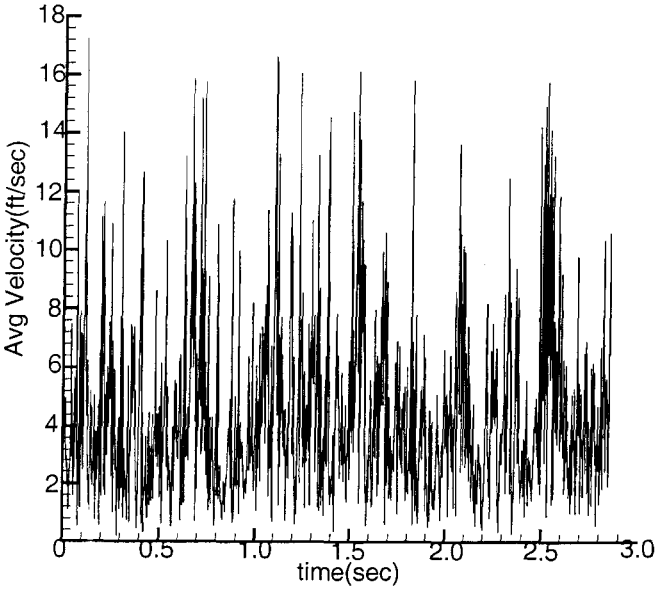
This implies that the change in the forces felt on the fuselage due to the wake structure in ground effect is quasi-steady. The rotorcraft would handle quite differently when compared to its handling out of ground effect at the same regime of flight. However, this change is a quasi-steady variation that can be trimmed off, if the changes are anticipated. All forces plotted are non-dimensionalized with rotor tip dynamic pressure at sea level standard density and fuselage

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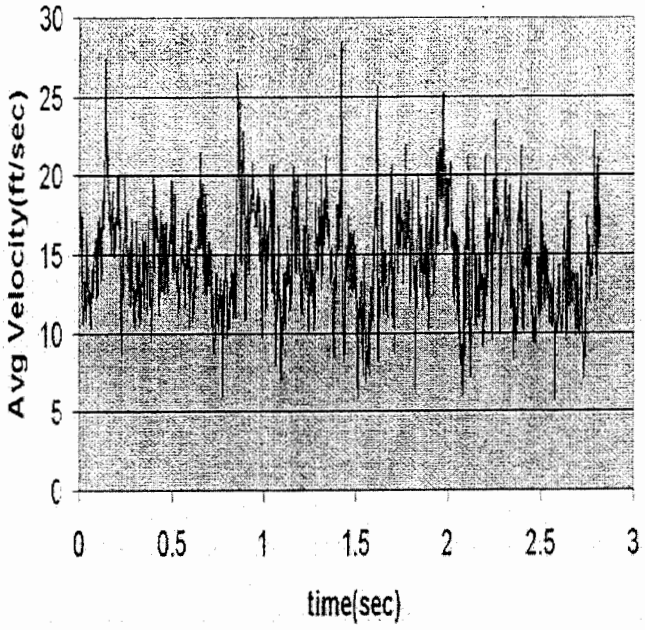
(a)



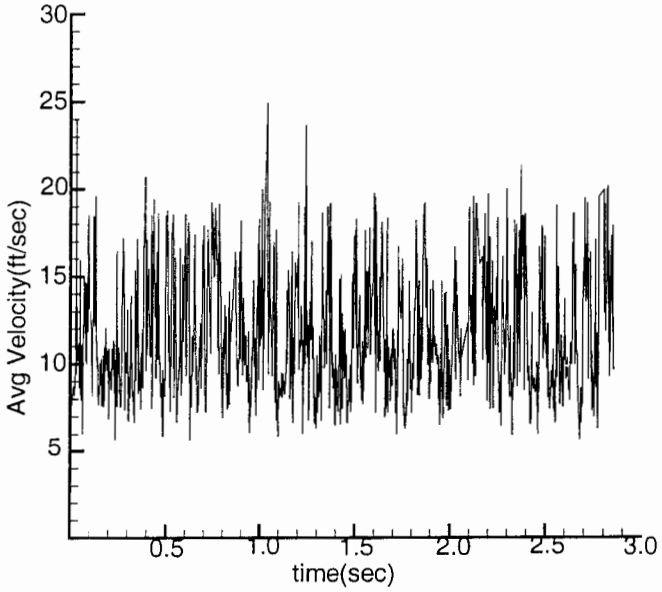
(b)

Figure 6. Comparison of experimentally and computationally obtained values of velocity for a two-bladed rotor for $h/R = 0.72$ for advance ratio $\mu = 0.03$ at Point P1. (a) Experimental Results (b) Computational results.

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(a)



(b)

Figure 7. Comparison of experimentally and computationally obtained values of velocity for a two-bladed rotor for $h/R = 0.72$ for advance ratio $\mu = 0.03$ at point P2. (a) Experimental results (b) Computational results.

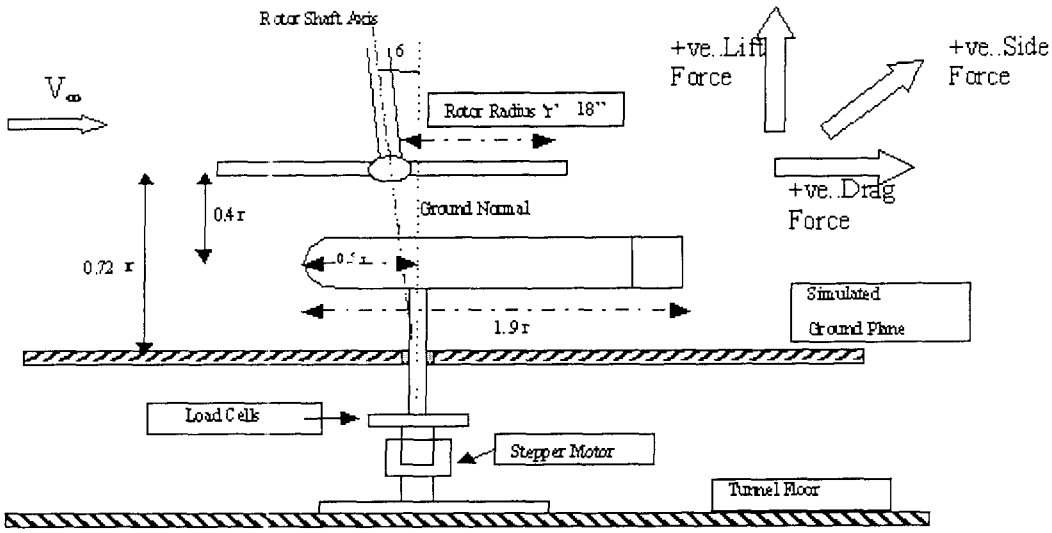


Figure 8. Experimental setup for fuselage force measurements.

planform area. The moment arms of the forces are non-dimensionalized using rotor radius with zero being the center of the load cell.

Another interesting phenomenon is a change in side force from positive to negative at μ of about 0.06. This is shown in Figure 9. Interestingly, this change in the direction of the side force occurs at the value of μ when there is a change from the recirculation regime and the ground vortex reaches the front of the fuselage. Furthermore, this change is associated with a change in slope of the curve, with a distinct kink at the μ corresponding to zero side force.

The moment arm of the side force is also plotted in Figure 9. These values were obtained by dividing the yawing moment by the side force, with the zero value being the center of the load cell as shown in Figure 9. The reversal of the side force is accompanied by a rearward shift of the side force moment arm. After that the IGE values approach the OGE values, as the ground vortex moves downstream. Beyond this μ value, the wake would interact with a tail rotor if present, causing a sudden change in tail rotor effectiveness as measured in Ref.9.

The variation of side-force with μ at various yaw angles is seen in Figure 10. It can be seen in Figure 10 that the yaw angle for zero side force changes with μ. Side force reverses at μ of about 0.05. This suggests a flow interaction rather than the effect of fuselage shape. Similarly, looking to the right along the zero side force line, we see that the zero side force at higher μ occurs at higher positive yaw angles.

a) Force vs. yaw OGE is symmetric about zero yaw. At zero yaw, side force becomes negative as soon as the regime of flight changes from hover to forward flight, and thereafter stays fairly constant with increasing μ. IGE, however, the side force variation is gradual and progressive at zero yaw with increasing μ. This can be seen by moving down the vertical dotted black line in Figure 11, and the resulting data are in Figure 9.

b) Side force magnitude at higher yaw is more in OGE than IGE.

c) An interesting phenomenon observed in ground effect is shown marked in the red dotted box in Figure 10. At μ above 0.06, side force increases suddenly between 20° and 60° yaw. This can be correlated with inputs obtained from various helicopter operators in the industry, who report considerable differences in the side forces produced IGE by fuselages of various cross sections.

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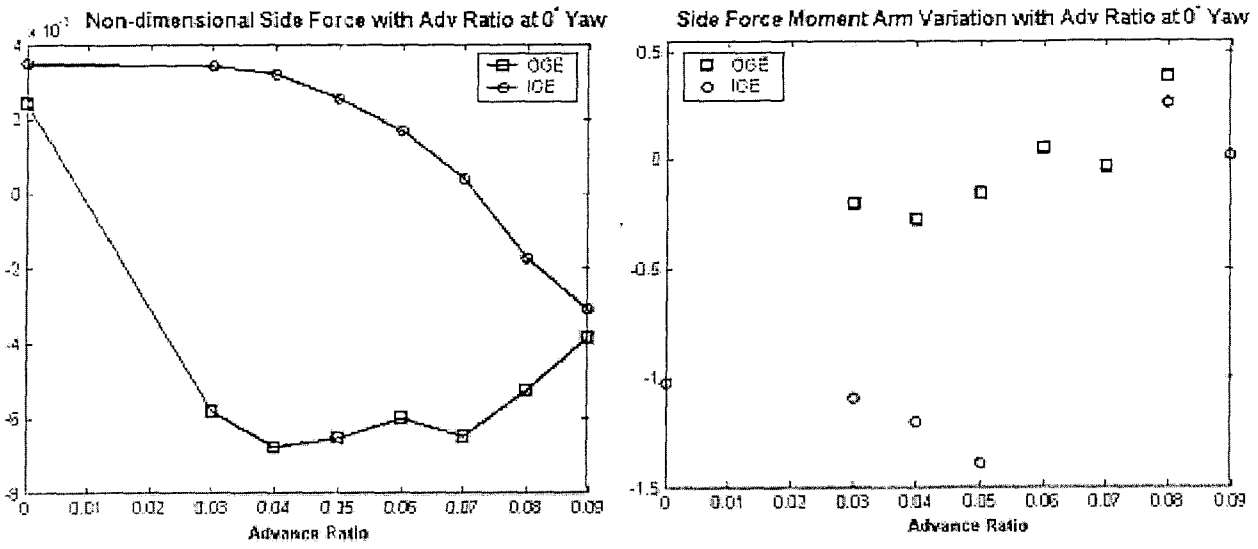


Figure 9. Variation of side force with advance ratio IGE.

Figure 11 compares fuselage download IGE and OGE, normalized using blade tip standard dynamic pressure and fuselage planform area.

a) Peak download coefficients (at zero yaw) reach 0.04. Given that the average downward velocity in the wake is roughly 10% of tip speed, this implies a fuselage cross-flow drag coefficient of 4 based on the velocity field. Much of this is obviously due to the blade pressure field impinging on the fuselage, causing very high force coefficients as seen in previous work on rotor-airframe aerodynamic interactions. Fuselage download IGE is far less, since the pressure pulses of blade passage reflect off the ground plane.

b) Between 20° and 60° yaw at μ above 0.07, there is a sudden reduction in IGE download, as in side force.

Figure 12 compares fuselage drag variations. OGE drag force increases consistently with μ.IGE, there is an increase in drag up to μ of 0.04, after which the drag decreases upto μ of 0.08, and then starts increasing again. Both plots have a vertical dotted line denoting 0 yaw, and a horizontal dotted line denoting 0 force.The data for the drag change at 0 yaw with μ is extracted and shown in Figure 13. The OGE value increases monotonically with advance ratio (but much slower than for a constant drag coefficient, while the IGE value stays constant till μ=0.04, falls to a minimum at μ=0.07, and rises again.The interesting point is that the drag reduction occurs at the point where the ground vortex reaches the front of the fuselage, and the rise occurs once the ground vortex is convected aft.

VII. Conclusions and Discussion

In previous work we have used flow visualization of a rotor in ground effect to capture two regimes of flight, with a recirculating flow pattern at an advance ratio less than 0.05, and a ground vortex regime between advance ratios of 0.05 to 0.08. Tip vortices were tracked, and it was found that the time scale of recirculation was about 0.11 seconds. This was correlated with the blips seen in the hotwire measurements in ground effect, corresponding to vortex interactions.

Computational results have been produced and compared with the experimental data. Specifically we have compared time traces of instantaneous magnitudes of the velocity at two points within the rotor wake. Qualitative agreement has been obtained; however there are quantitative differences. The source of the discrepancies may be due to two features of the numerical computations. These are; the computations were done for lifting line while the experiments

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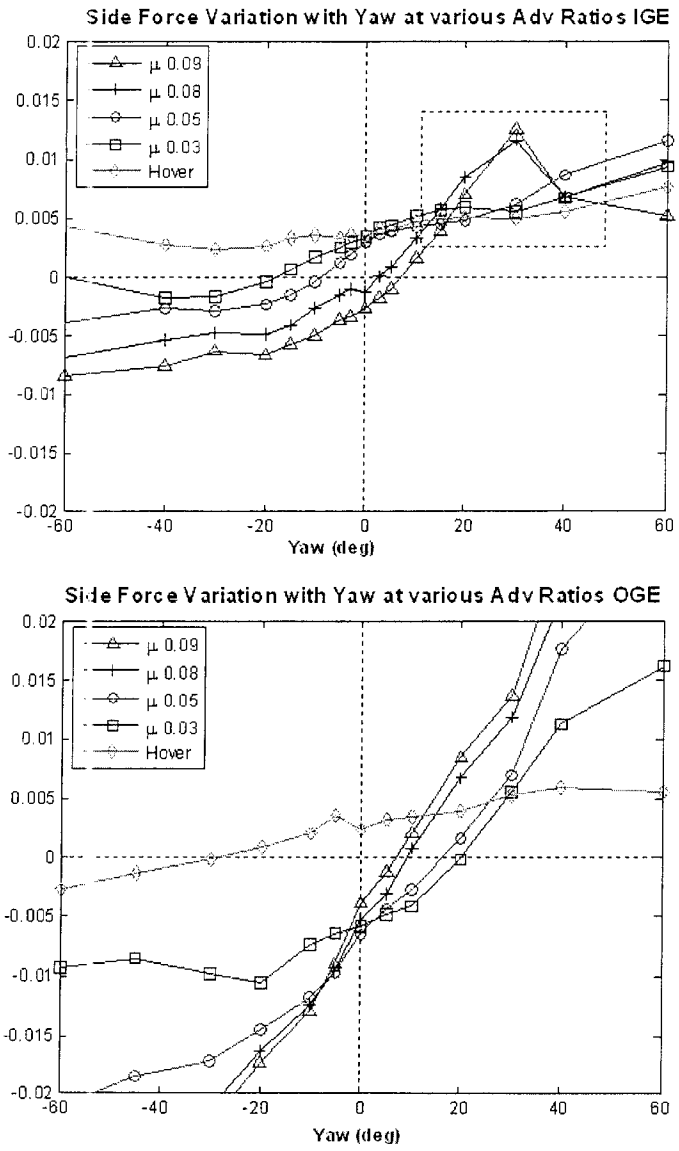


Figure 10. Variation of side force with yaw angle at various advance ratios. Values are normalized using blade tip standard dynamic pressure and fuselage planform area.

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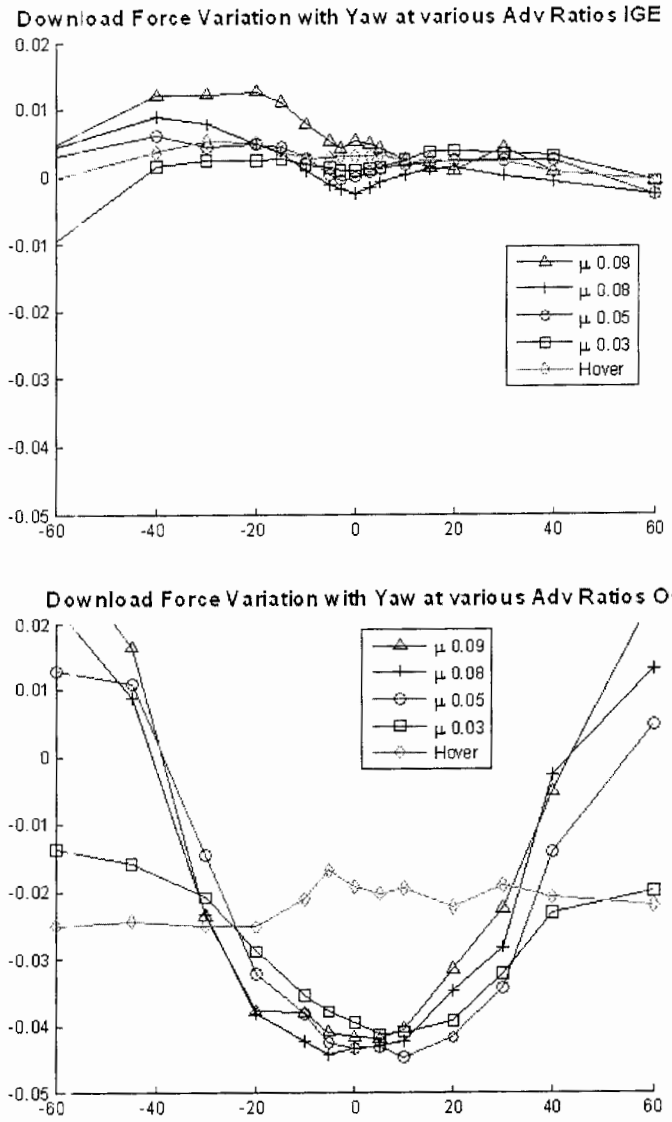


Figure 11. Comparison of fuselage download force IGE vs. OGE. Values are normalized using blade tip standard dynamic pressure and fuselage planform area.

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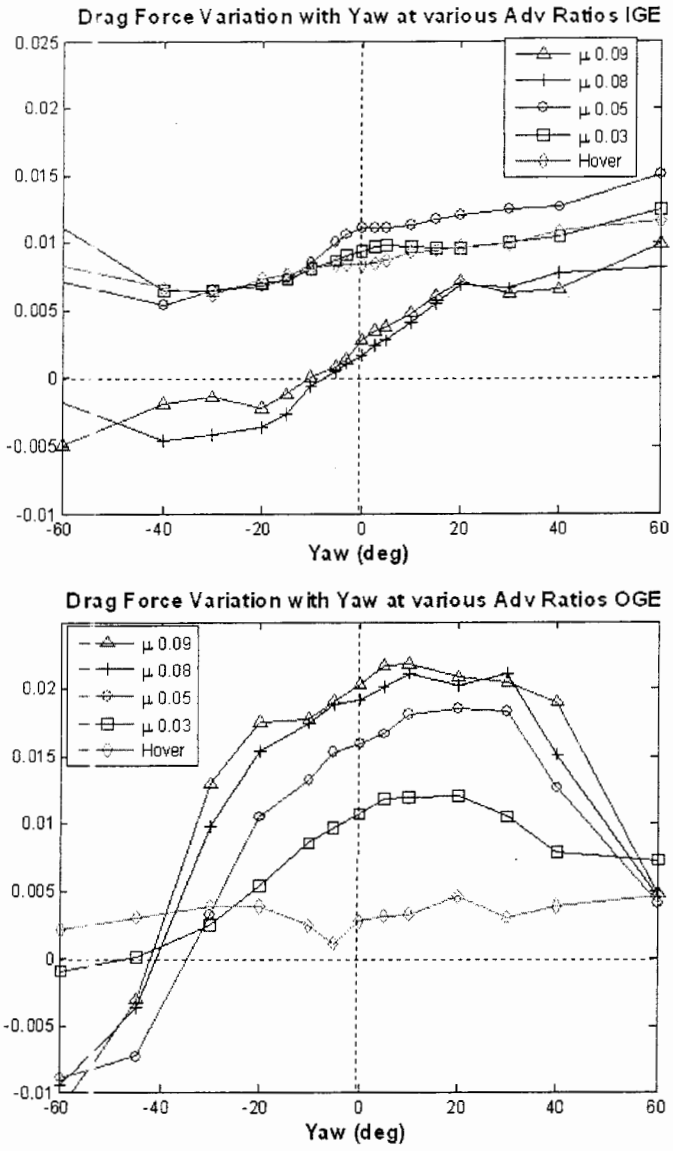


Figure 12. Comparison of fuselage drag IGE vs. OGE. Values are normalized using blade tip standard dynamic pressure and fuselage planform area.

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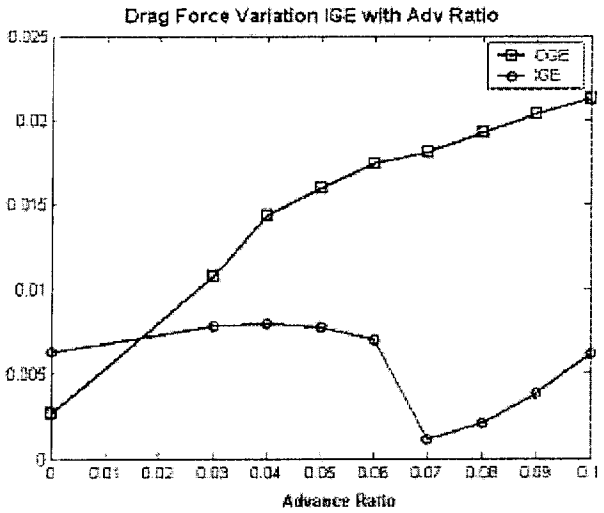


Figure 13. Comparison of fuselage drag IGE vs. OGE at 0 Yaw. Values are normalized using blade tip standard dynamic pressure and fuselage planform area .

are performed with blades having a span to chord ratio of 5. Second, it is difficult to determine the beginning of a different cycle in the computations. The results shown here are velocity traces between revolutions 50 and 150. Even so qualitative agreement with experiment is obtained. Both of these issues are presently under investigation.

New experiments have used a fuselage model to correlate fuselage loads with flowfield data. New conclusions are listed below.

1. A peak frequency component of 1 Hz was seen in the IGE hotwire data - with the fundamental blade passage frequency being 35Hz. Inherent unsteadiness in the flow pattern due to a rotor in ground effect without a fuselage, has been investigated and found in flow visualization and hotwire anemometry. This situation is fundamentally unsteady, and requires long-period unsteady simulation methods with high accuracy. These long time scales pose a major challenge for numerical predictions, but appear to be due to vortex interactions, which can be modeled by inviscid methods. They are postulated to cause rotor thrust and torque variations, as well as control changes due to ingestion of the main rotor wake into the tail rotor. This could cause loss in tail rotor thrust, as studied in Ref.9.

2. Fuselage loads did not show low-frequency fluctuations, and instead exhibited loads at the fundamental (i.e., wake loads) and first harmonic (i.e., blade passage) of the rotor frequency. Long-time-scale flowfield unsteadiness is integrated out along the fuselage. Thus fuselage loads are postulated to be predictable using quasi-steady-state prediction methods, as changes that occur between different flow states.

3. Measurements indicate a reversal of the sideforce as a helicopter accelerates in ground effect, and this switch seems to occur at the advance ratio where the ground vortex position reaches the fuselage. A sudden increase in side force at positive yaw angles at advance ratios above 0.06 IGE. Considerable reduction in the fuselage download force IGE is attributed to change in wake structure. Fuselage drag decreases during encounter with the ground vortex.

Experimental uncertainity has also been addressed in this work. A key feature of this experiment is that the flow in the OGE case has been captured with sufficient low uncertainty that the IGE fluctuations can be properly quantified. This issue is examined using flow visualization, hot-wire measurements and now using strain gage / load

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cell measurements of fuselage loads. Error in quantifying ground vortex position with respect to is examined below. The position was plotted by digitizing video images, and marking the edge of the void area that was formed due to lack of seed particles in the vortex. The coverage area of the image was 60 inches by 25 inches. The image was digitized into a 1280 by 768 pixels bitmap. Since the edge of the void area cannot be resolved under one pixel, the error probability of the position is approximately 0.04 in, or 0.0022 R, or 10% of typical vortex core size. Position accuracy of the hot-wire sensor location, as aligned using a laser beam and checked with test runs (Ref. 19) , is no more than 1mm. The wire alignment with the y-axis is better than 0.1 degree. The force balance was calibrated with precision weights, and the response was found to be linear. Five sets of check runs were done with precision weights, both in the negative and positive side force direction. The error was found to be less than 1% for the side force. The 6-d.o.f load cell being used has an error less than 0.5% of full load. The force error is on the order of 0.01 lbf with typical side force and drag values on the order of 1lbf.

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